

AYMANE AIT DADS

Algorithm Developer Intern | Data Science · ML/AI · Sensor Signal Analytics

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PROFESSIONAL SUMMARY

Built and deployed an end-to-end machine learning pipeline on 100K+ semiconductor-grade network records, combining Random Forest classification and K-Means clustering to map performance profiles across thousands of nodes with a 60% reduction in manual analysis time. Achieved top 10% on the NVIDIA Nemotron Kaggle leaderboard (0.80/1.0, \$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE model using LoRA with ~888M trainable parameters across 20+ single-variable ablation studies. Proficient in Python, Scikit-learn, PyTorch, Pandas, feature engineering, and advanced data analytics applied to large-scale sensor signal datasets. Available for a full-time internship starting Summer 2025; currently based in France and open to relocation to Santa Clara, CA (visa sponsorship required).

CORE COMPETENCIES

Data Science | Machine Learning | Sensor Signal Analytics | Feature Engineering | Advanced Data Analytics | Algorithm Development | Process Optimization | Workflow Acceleration

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and deployed an end-to-end machine learning pipeline on **100K+ fiber-optic network sensor records** (upload speed, latency, jitter) using Random Forest classification and K-Means clustering to assess network quality across thousands of nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for maintenance scheduling decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, cross-validated model evaluation, and reproducible data preprocessing workflows built in Python and Scikit-learn.
- Delivered stakeholder presentations translating **algorithm outputs into actionable maintenance recommendations**, visualized through Power BI dashboards for non-technical process engineers and business leads.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score of 0.80/1.0** (\$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought logic examples using SFTTrainer completion-only loss.
- Resolved 5 Blackwell GPU (sm_120) incompatibilities via monkey-patching, kernel stubbing, and PyTorch fallbacks, and conducted **20+ single-variable ablation studies** confirming packing, synthetic data, and adapter-size as critical performance levers.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · Predictive Maintenance

- Engineered an unsupervised fault-detection system for slide-rail machinery using a Transformer autoencoder on **Mel spectrogram sensor signals with SpecAugment**, achieving AUC 0.80 without any labeled anomaly examples.
- Applied advanced data analytics to raw time-series sensor data, demonstrating algorithm development for industrial equipment monitoring directly analogous to **Etch and semiconductor process tool signal analysis**.

PyTorch · Librosa · NumPy · Scikit-learn

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: Python | PyTorch | Scikit-learn | Hugging Face Transformers | TensorFlow | PEFT | Unsloth

Data Science & Analytics: Pandas | NumPy | Feature Engineering | Clustering | Ensemble Methods | Power BI | SQL

MLOps & Infrastructure: Git | Linux | FP16 Mixed Precision | CUDA Debugging | Ablation Studies | Docker | Bash